

Lab Program 3

Code a Dockerized python flask or Node.js application

Name : Harshavardhan B R

USN : 1RV24MC042

Step 1 : Create required files and folders

```
> ls
app.py  Dockerfile  requirements.txt
> tree
.
├── app.py
├── Dockerfile
└── requirements.txt

1 directory, 3 files
```

Step 2: Add necessary content

```
//app.py
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello world"
if __name__ == "__main__":
    app.run("0.0.0.0",3000)
```

```
#requirements.txt
Flask
```

```
FROM python:latest

WORKDIR /app

COPY . .

RUN pip install --no-cache-dir -r requirements.txt

EXPOSE 3000

CMD [ "python", "app.py" ]
```

Step 3: Build a docker image

```
> docker build -t thirdprog .
> docker images | grep thirdprog
```

```

1 directory, 5 files
> docker build -t thirdprog .
[+] Building 3.2s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 170B                               0.0s
=> [internal] load metadata for docker.io/library/python:late     3.1s
=> [auth] library/python:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/4] FROM docker.io/library/python:latest@sha256:1ad1a43b    0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 93B                                    0.0s
=> CACHED [2/4] WORKDIR /app                                       0.0s
=> CACHED [3/4] COPY . .                                           0.0s
=> CACHED [4/4] RUN pip install --no-cache-dir -r requirement    0.0s
=> exporting to image                                              0.0s
=> => exporting layers                                             0.0s
=> => writing image sha256:c87b00c4379a2f109ecd74923cdc023684    0.0s
=> => naming to docker.io/library/thirdprog                       0.0s

```

```

> docker images | grep thirdprog
thirdprog          latest          c87b00c4379a    25 minutes ago    1.13GB

```

Step 4 : Run the container

```
> docker run -p 6000:3000 thirdprog
```

```

❯ > docker run -p 5000:3000 thirdprog
docker: Error response from daemon: failed to set up container networking: driver failed
programming external connectivity on endpoint determined_bose (bf7fcedf96a14bbf9a6a3ff8
e684b6d5ceafc66f6456bd657cd9e4d954ca7e5f): Bind for 0.0.0.0:5000 failed: port is already
allocated

Run 'docker run --help' for more information
❯ > docker run -p 6000:3000 thirdprog
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a p
roduction WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3000
* Running on http://172.17.0.2:3000
Press CTRL+C to quit

```

Step 5 : Check and verify

```

❯ curl -X GET localhost:6000
Hello world!

```

```

❯ > docker ps | grep thirdprog
befbc4ef8f14      thirdprog        "python app.py"        2 minutes ago    Up 2 minutes      0.0.0.0:6000->3000/tcp, [::]:6000->3000/tcp    nice_taussig

```